What past home values predicted

Week 1 historical forecasts | Los Angeles County

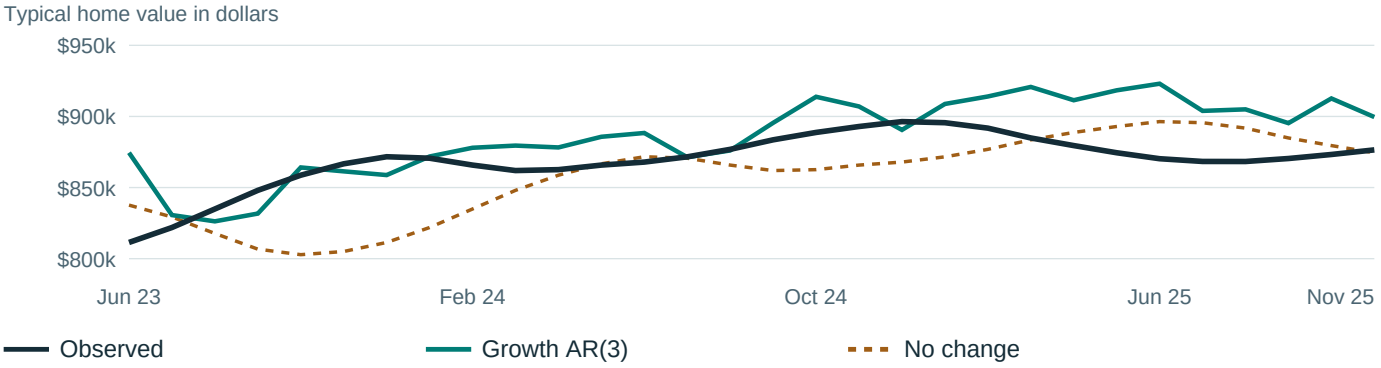
1. How much of future housing-market behavior can be predicted from the variable's own past?

Past prices helped over short horizons. Growth AR(6) reduced three-month test RMSE by 36.5% relative to holding the last home value constant. **Holding the last home value constant beat the validation-selected models on test RMSE at all three horizons.** We tested 30 monthly forecast origins from 2022-12 through 2025-05.

Independent test against the no-change benchmark

Horizon	Validation choice	Test MAE	Test RMSE	No-change RMSE
3 months	Level AR(1)	$16,360	$20,147	$18,241
6 months	Level AR(1)	$25,372	$30,160	$27,766
12 months	Growth AR(1)	$46,040	$54,856	$31,587

Six-month forecasts from the three-lag growth model



2. How sensitive are the results to lag length, training period, and prediction horizon?

At the first validation origin (2020-12), Growth AR(6) had the smallest training error (0.313 percentage points, 65 fitted months). AR(6) also had the lowest test RMSE of the three growth models at every horizon. That retrospective ranking uses held-out outcomes. The choices in the table used only the earlier validation period.

Starting the AR(3) sample in 2010, 2015, 2019 gave first-origin lag-one coefficients of 1.49, 1.47, 1.68, respectively. Six-month test RMSE was $23,469, $25,911, $27,720. Later starts give pandemic-era price changes more weight and leave fewer months to estimate persistence. An expanding window adds a month at each origin. A rolling window also moves the start.

At six months, AR(1) in price levels had RMSE of $30,160, versus $35,224 for AR(1) in growth and $26,773 for ARIMA(0,1,1) on log levels.

Horizon mattered strongly. Growth AR(6) test RMSE rose from $11,590 at three months to $46,323 at twelve months, exceeding the twelve-month no-change error of $31,587.

How forecasts varied by market

Common target months | Six-month horizon

3. Do the same simple models work across home prices, rents, permits, inventory, and California counties?

The same autoregressive method could be applied to every available outcome and county. Its accuracy varied with the market, outcome transformation and horizon, so a common model did not give a uniform forecasting advantage.

California counties

County	AR(1) RMSE	AR(3) RMSE	AR(3) MAPE	Growth SD
Los Angeles	$35,224	$25,911	2.41%	0.65 pp
Orange	$35,234	$27,428	2.12%	0.60 pp
Riverside	$13,529	$12,075	1.69%	0.37 pp
San Diego	$28,765	$25,857	2.36%	0.60 pp

All 4 counties had observations at every target month. AR(3) beat AR(1) on six-month RMSE in 4 of them. At twelve months AR(1) beat AR(3) in Riverside. Los Angeles had both the highest AR(3) mean absolute percentage error (MAPE) and the most variable monthly price growth. Growth SD measures monthly log-change volatility in percentage points. A statewide ZHVI series was unavailable in the local database.

Rents, permits and homes for sale

Rent. On identical six-month dates AR(3) missed Zillow's smoothed, seasonally adjusted rent index by 0.83% on average, versus 2.41% for home values. Growth AR(1) had the lowest test RMSE of the three growth models at all horizons.

Permits. On the same trailing three-month mean of permitted units, test RMSE was 277 for AR(1) on the mean, 276 for AR(1) on raw permits followed by averaging, and 362 for seasonal naive. Adjacent means share two months, which creates persistence mechanically. Census flags many published permit estimates as imputed. Monthly survey coverage expanded in 2022.

Inventory. Six-month RMSE was 3,522 homes for seasonal naive and 2,777 for level AR(3). Level AR(1) beat the other level model at every horizon. Through the first test origin, calendar month accounted for 87% of monthly log-change variation in unadjusted inventory versus 6% in Zillow's already smoothed and seasonally adjusted home values. The benchmark uses the level twelve months before the target.

How to read the backtest

Validation used origins 2020-12 to 2021-12. Models use their own predictions when future lagged values are needed. Error is forecast minus actual. MAE averages absolute errors. RMSE gives large misses more weight.

Provider histories in the current DuckDB incorporate revisions and smoothing. These retrospective tests do not reconstruct what was published at each origin. Monthly origins also yield overlapping multi-month errors. Results apply to these counties, outcomes and dates. The CSV tables retain the sample months, coefficients and errors.